Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/26/2024
PI Name:	Williams, Michael A. M.D.		
Project Title:	Pilot Feasibility Study of Safety and Effectiveness of Telemetric Intracranial Pressure Monitoring in Astronauts Before, During, and After Commercial Orbital Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	11/01/2022	**End Date:**	02/29/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-EXP0014		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Spaceflight Associated Neuro-ocular Syndrome (SANS) is a disorder after long-duration spaceflight characterized by swelling of the optic nerve (papilledema) folds in the choroid layer of the eye, flattening of the optic globe of the eye, and change in near vision (Mader, 2011). Up to 70% of astronauts develop SANS, which is considered a “red risk” – because if severe visual impairment developed, astronauts’ ability to function safely or to control a spacecraft could be compromised (Patel ZS, 2020). SANS is thought to be caused by abnormal intracranial pressure (ICP); however non-invasive ICP methods researched by NASA have not been accurate enough to make this determination (Stenger MR, 2017). Lumbar punctures are considered too risky to perform during spaceflight (Barr YR, 2014). The underlying cause of SANS remains unknown, and, despite evidence during parabolic flight that ICP may not be pathologically elevated during brief periods of weightlessness (Lawley, 2017), measures of ICP during long-duration spaceflight are needed to understand whether abnormal ICP plays a role in the development of ocular and brain structural changes. Accordingly,		

	this proposal is relevant to mitigating the gap SANS-102: Determine the relationship between the fluid shifts induced ocular changes and fluid shifts in the central nervous system (CNS), including whether elevated intracranial pressure or brain edema play a role.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	IMPACT: The successful completion of the FDA QSub and IDE EFS applications provides regulatory approval for surgical implantation of the Miethke M.scio telemetric ICP sensor, which will demonstrate that ICP can be measured accurately and safely before, during, and after spaceflight. If the pilot early feasibility study is successful, then future research using the same FDA IDE EFS approval can be submitted to the FDA.
Task Progress:	KEY FINDINGS: The UW investigators worked intensely with collaborators at Miethke, University of Umea, University of Pennsylvania, and Baylor College of Medicine, as well as with TRISH leadership and key SpaceX personnel to reach all objectives. The following deliverables were produced. All are consistent with the description and timeline in the proposal. • FDA QSub submitted April 18, 2022 • Teleconference with FDA QSub reviewers on July 5, 2022 resulted in their requirement for additional testing of the Miethke equipment that would have taken 6-12 months and cost an additional $1million. • Teleconference with FDA/NASA liaisons (FDA CDRH Countermeasures Office), and TRISH and NASA leaders on August 19, 2022 resulted in a request to meet with FDA reviewers to reconsider the IDE as an early feasibility study (EFS), which would forego the need for additional testing of the Miethke equipment. • Teleconference with FDA reviewers on October 24, 2022 resulted in FDA approval to proceed with IDE EFS application. • Grant proposal EXP-2022-0000000014, titled "Pilot Feasibility Study of Safety and Effectiveness of Telemetric Intracranial Pressure Monitoring in Astronauts Before, During, and After Commercial Orbital Spaceflight" submitted to TRISH August 9, 2022. The proposal plans for the implantation surgery of the M.scio assembly to be performed at the University of Washington rather than in the UK. Determination of UK regulatory requirements was no longer needed. • Banbury Center Conference "Developing an Ethical Framework for Human Research in Private Spaceflight", December 9-12, 2022. This conference was a direct outcome of early discussions of this project with TRISH leadership and results in submission of a policy manuscript to Science. • FDA IDE EFS application with research protocol and safety assessments submitted December 29, 2022 and, following 5 revisions through interactive review with the FDA, was approved February 14, 2023 • Baylor IRB application and approval March 16, 2023 • Space flight participant consent for the research protocol obtained March 22, 2023 • Experiment selected for flight May 17, 2023 ---------- The experience with the lumbar catheter-M.scio implant, and recommendations for changes to the protocol based on the final Root Cause Analysis report, will guide future investigations using this approach. The risk/benefit analysis based on our experience can be compared to expected risk/benefit analysis for other approaches that might be considered, such as implantation of an intracranial device. The FDA Early Feasibility Studies (EFS) Investigational Device Exemption (IDE) remains active with the Principal Investigator (PI) as sponsor-investigator.
Bibliography Type:	Description: (Last Updated: 05/15/2026)
Articles in Peer-reviewed Journals	Rahimzadeh V, Fogarty J, Caulfield T, Auñón-Chancellor S, Borry P, Candia J, Cohen IG, Covington M, Fernandez Lynch H, Greely HT, Hanlon M, Hatt J, Low L, Menikoff J, Meslin EM, Platts S, Ravitsky V, Ruttley T, Seidler RD, Sugarman J, Urquieta E, Williams MA, Wolpe PR, Donoviel D, McGuire AL. "Ethically cleared to launch?" Science. 2023 Sep;381(6665):1408-11. https://www.science.org/doi/abs/10.1126/science.adh9028 ; PMID: 37769066 , Sep-2023